Machine Learning Based Buyer Segmentation and Investment Profiling for Real Estate Market Intelligence.

Eng. Yunes Abdulghani Mohammed Ghaleb

AI & ML Engineer,
Unified Mentor Project

alshameeri.ai.eng@gmail.com
+91 7338167297
+967 773792775

1. Abstract

This paper presents an end-to-end machine learning solution for segmenting real estate buyers using unsupervised learning techniques. The system processes 2,000 client records across demographic, financial, and geographic dimensions to discover four distinct buyer segments: Global Investors (77.3%), First-Time Buyers (20.5%), Corporate Buyers (1.3%), and Luxury Investors (1.0%). K-Means clustering was employed as the primary algorithm, validated through Hierarchical Clustering, Elbow Method, Silhouette Score, and Davies-Bouldin Index. The resulting segments were deployed through an interactive Streamlit dashboard enabling real-time filtering, deep-dive analytics, geographic visualization, and AI-powered segment prediction for new clients. Business recommendations tailored to each segment demonstrate clear pathways for improved customer targeting, marketing efficiency, and investment decision support.

Keywords: Buyer Segmentation, K-Means Clustering, Real Estate Analytics, Business Intelligence, Streamlit, Unsupervised Learning.

2. Introduction

2.1 Background

The global real estate market serves an increasingly diverse customer base spanning individual homeowners, institutional investors, international buyers, high-net-worth individuals, and first-time purchasers. Traditional customer relationship management approaches treat all buyers uniformly, leading to inefficient resource allocation, generic marketing campaigns, and missed revenue opportunities. The emergence of data-driven customer segmentation offers a paradigm shift, enabling companies to identify latent patterns in buyer behavior and tailor strategies accordingly.

PARCL Co. Limited operates within this competitive landscape, managing thousands of client relationships across multiple countries and regions. Without systematic segmentation, the company lacks the analytical foundation to differentiate between a high-volume institutional investor and a first-time residential buyer, despite their vastly different needs, risk profiles, and lifetime value potentials.

2.2 Problem Statement

PARCL currently lacks a data-driven understanding of:

Different types of property buyers and their distinct characteristics

Investment motivations across demographic and geographic segments

Geographic differences in investment behavior and preferences

Customer financing patterns and loan application trends

Satisfaction drivers and their relationship to segment membership

This knowledge gap manifests in three critical business consequences:

Inefficient marketing spend — campaigns are broadcast uniformly without segment-specific messaging

Poor customer targeting — high-value opportunities are not prioritized

Missed investment opportunities — geographic and behavioral trends remain undetected

2.3 Objectives

This project seeks to:

Build an intelligent buyer segmentation system using unsupervised machine learning

Identify and characterize high-value customer groups, Analyze investment patterns across segments

Visualize customer distributions geographically and behaviorally

Generate actionable business insights for strategic planning

Deploy an interactive dashboard for ongoing business intelligence

Enable AI-powered prediction of buyer segments for new client acquisition.

2.4 Scope & Limitations

Scope: The analysis covers 2,000 client records from multiple countries (USA, UK, Canada, Germany, France, Australia, Russia, Mexico, Belgium, Denmark) with 12+ features spanning demographics, investment behavior, property characteristics, and customer satisfaction.

Limitations:

The dataset is static (snapshot in time) rather than longitudinal

Causal relationships between segments and outcomes cannot be established without controlled experiments

External market factors (interest rates, economic cycles) are not incorporated

The model assumes cluster stability; retraining is required as market conditions evolve

3. Dataset Description

3.1 Data Source

The dataset comprises two primary tables:

clients.csv — Client demographic and behavioral attributes

properties.csv — Property transaction details linked to clients

These were merged and processed to create processed_data.csv, the analytical foundation for segmentation.

🔍 Data Preview

	client_id	age	gender	client_type	country	region	acquisition_purpose	loan_applied	referral_channel	satisfaction_score	avg_sale_price	total_investment	property_count	avg_floor_area	total_transa
0	C0001	58	F	Individual	Usa	California	Home	Yes	Website	4	311691.18	1246764.72	4	983.885	
1	C0002	64	M	Individual	Usa	California	Home	No	Website	1	368219.186	1841095.93	5	1187.942	
2	C0003	67	M	Individual	Usa	California	Home	Yes	Agency	4	332291.518	1661457.59	5	1058.11	
3	C0004	67	M	Individual	Usa	California	Home	No	Website	5	268043.9183	1608263.51	6	937.1033	
5	C0006	69	M	Individual	Usa	California	Home	Yes	Website	3	302826.212	1514131.06	5	989.59	
6	C0007	79	M	Individual	Canada	Quebec	Investment	No	Website	5	295558.8936	3251147.83	11	995.6936	
7	C0008	57	F	Individual	Usa	California	Home	No	Website	5	295346.082	1476730.41	5	1031.106	
8	C0009	51	M	Individual	Usa	California	Investment	No	Agency	5	364166.47	1820832.35	5	1305.644	
9	C0010	60	M	Individual	Usa	Oregon	Investment	No	Agency	3	346486.8	1732434	5	1140.206	
10	C0011	70	F	Individual	Usa	California	Home	Yes	Website	2	265042.304	1325211.52	5	935.914	

Figure 1: Sample Records from the Real Estate Buyer Dataset Showing Demographic, Financial, and Property Attributes

3.2 Feature Dictionary

Feature	Type	Description
client_id	String	Unique client identifier (C0001–C2000)
client_type	Categorical	Individual or Corporate
gender	Categorical	Male (M) or Female (F)
country	Categorical	Country of residence
region	Categorical	Geographic region/state/province
date_of_birth	Date	Birth date (derived to age)
acquisition_purpose	Categorical	Investment or Home (personal use)
loan_applied	Binary	Yes/No financing indicator
referral_channel	Categorical	Website, Agency, Referral, etc.
satisfaction_score	Ordinal (1–5)	Customer satisfaction rating
avg_sale_price	Continuous	Average property purchase price (USD)
total_investment	Continuous	Total investment value (USD)
property_count	Integer	Number of properties owned
floor_area_sqft	Continuous	Total floor area in square feet

3.3 Descriptive Statistics

Metric	Value
Total Records	1,999 (99.95% of 2,000; 1 record removed during cleaning)
Countries	10
Regions	20+
Average Age	54.8 years
Average Investment	$1,259,200
Average Satisfaction	3.03 / 5.0
Average Properties	3.6 per client
Loan Application Rate	36.8%

4. Methodology

4.1 Data Preprocessing

The preprocessing pipeline executed the following operations:

Missing Value Treatment: Identified and handled missing client attributes through imputation and record removal (1 record excluded)

Duplicate Elimination: Removed duplicate client entries to prevent cluster bias

Label Normalization: Standardized categorical labels (e.g., "USA" vs. "Usa" unified)

Age Derivation: Converted date_of_birth to age in years

Outlier Assessment: Evaluated extreme values in investment and property counts; retained as they represent legitimate high-value segments

4.2 Exploratory Data Analysis (EDA)

EDA revealed several critical patterns:

Geographic Concentration: USA (particularly California) dominates the client base, representing the majority of high-investment activity

Investment Skew: Total investment ranges from ~$500K to ~$3.5M, with significant right skew

Age Distribution: Buyers span 25–95 years with peaks in the 45–65 range

Purpose Split: Approximately 70% Home, 30% Investment across the dataset

Loan Patterns: First-Time Buyers show higher loan dependency; Global Investors predominantly self-finance.

4.3 Feature Engineering

Categorical variables were encoded using:

Label Encoding: For ordinal or binary variables (gender, loan_applied)

One-Hot Encoding: For nominal variables with multiple categories (client_type, country, region, acquisition_purpose, referral_channel)

Numeric variables were standardized using StandardScaler to ensure equal contribution to distance-based clustering:

age, satisfaction_score, avg_sale_price, total_investment ,property_count and floor_area_sqft

Feature selection prioritized variables with demonstrated discriminative power for buyer behavior differentiation.

4.4 Clustering Algorithms

4.4.1 K-Means Clustering

K-Means was selected as the primary algorithm due to:

Computational efficiency: O(n × k × i × d) complexity suitable for 2,000 records

Interpretability: Clear centroid-based segment definitions

Scalability: Easy retraining as new client data arrives

The algorithm was configured with:

k = 4 (determined via Elbow Method)

init = 'k-means++' for optimal centroid initialization

max_iter = 300

n_init = 10 for robustness

To better understand the separation among the identified buyer segments, Principal Component Analysis (PCA) was applied to project the high-dimensional feature space into two principal components, as illustrated in Figure 2.

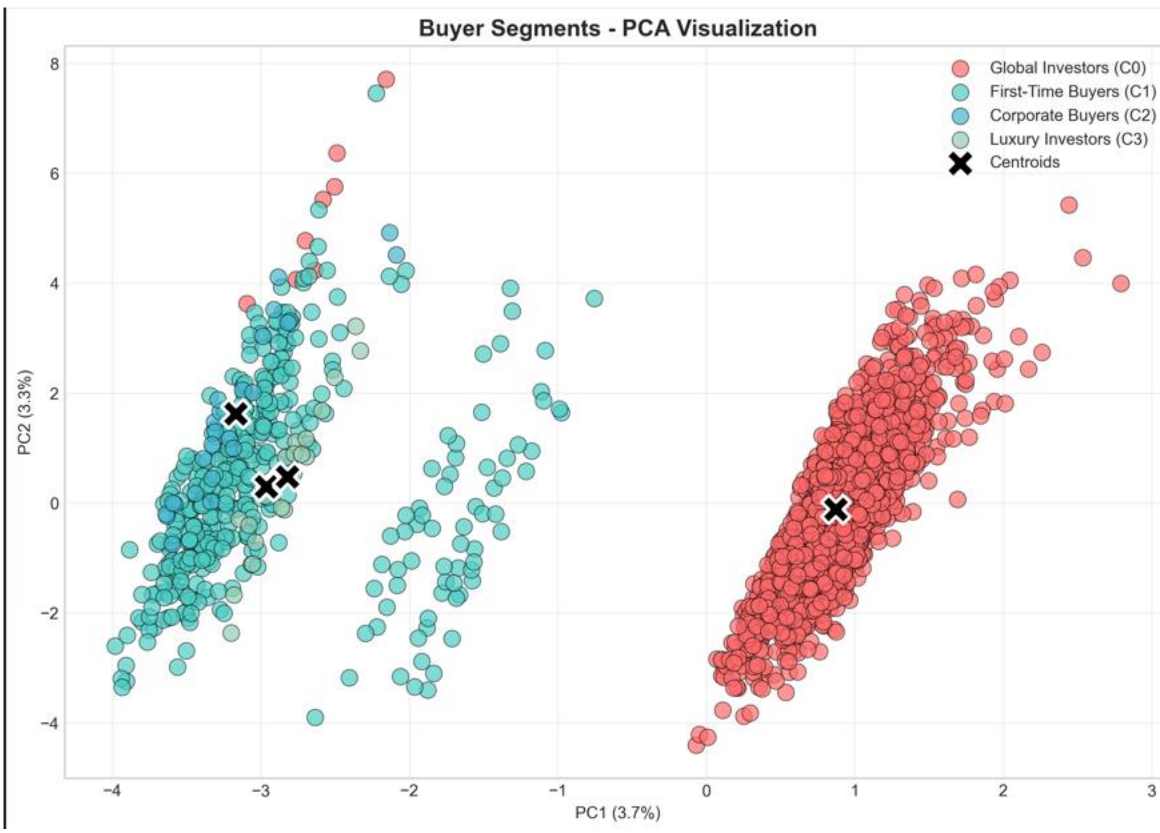

Figure 2. PCA Projection of the Identified Buyer Segments

4.4.2 Hierarchical Clustering

Agglomerative Hierarchical Clustering was employed as a validation mechanism:

Linkage method: Ward's method (minimizes within-cluster variance)

Distance metric: Euclidean

Purpose: Validates K-Means results by revealing nested cluster relationships and confirming the 4-cluster structure through dendrogram analysis

The dendrogram supported the 4-cluster solution, with clear separation between the dominant Global Investors segment and the smaller specialized segments.

Hierarchical clustering was additionally employed to validate the segmentation obtained by K-Means. The resulting dendrogram illustrates the hierarchical relationships among buyer groups, as shown in Figure 3.

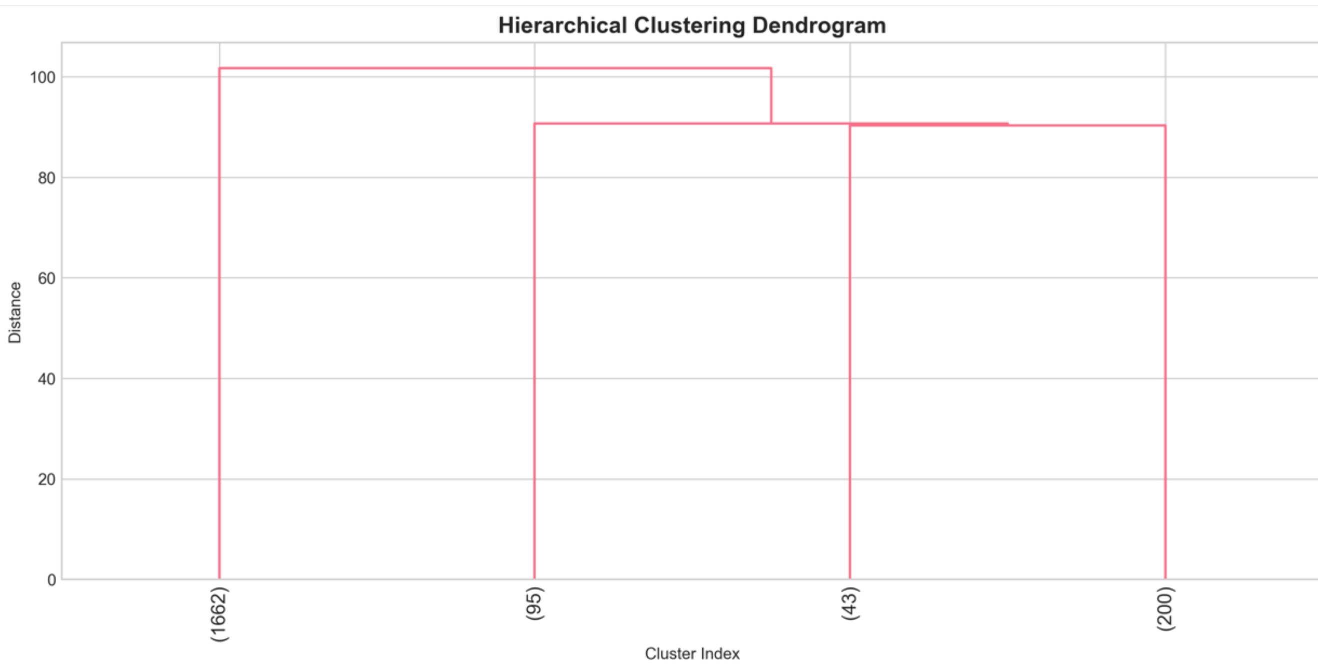

Figure 3. Hierarchical Clustering Dendrogram Showing Buyer Relationships.

4.5 Model Evaluation & Validation

4.5.1 Elbow Method

The Elbow Method was applied across k = 2 to 10. The inertia curve exhibited a distinct elbow at k = 4, indicating optimal trade-off between model complexity and within-cluster sum of squares.

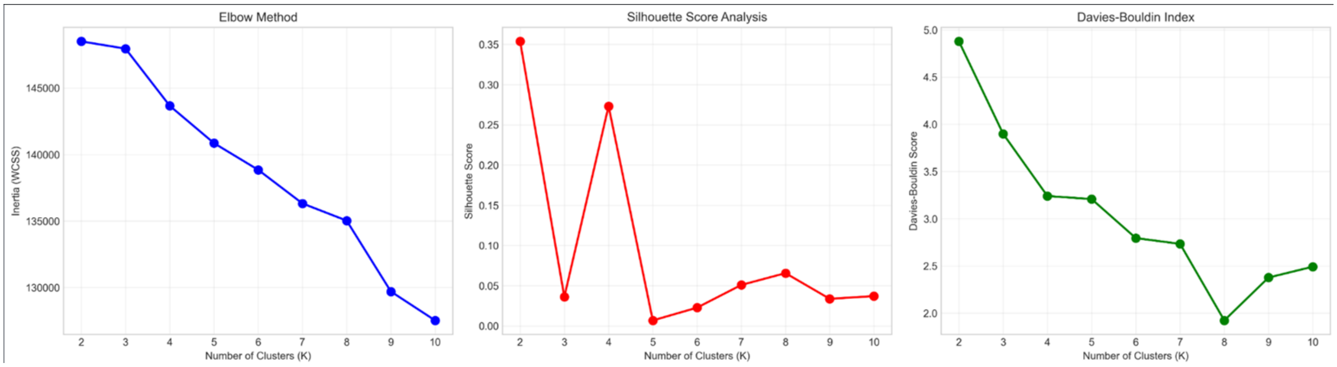

Figure 4. Cluster Validation Results using the Elbow Method, Silhouette Score, and Davies–Bouldin Index.

5. Results & Analysis

5.1 Cluster Profiles

Segment	Label	Count	Share	Avg Age	Avg Investment	Avg Satisfaction	Avg Properties	Loan Rate	Dominant Purpose
C1	Global Investors	1,545	77.3%	56.5	$1,269,189	3.03	3.67	~37%	Home
C2	First-Time Buyers	409	20.5%	54.8	$1,214,189	3.05	3.55	~62%	Home
C3	Corporate Buyers	25	1.3%	53.5	$1,365,332	2.80	3.88	~16%	Investment
C4	Luxury Investors	20	1.0%	55.8	$1,273,197	3.05	3.65	~15%	Home

Key Insight: Despite similar average investment values, segments differ dramatically in loan behavior, property volume, and satisfaction drivers.

5.2 Geographic Analysis

USA Dominance: California, Texas, Florida, New York represent the highest client concentrations and investment volumes.

International Distribution: UK (England, Scotland), Canada (Ontario, Quebec), Germany (Bavaria), Australia (Victoria) show moderate activity.

Emerging Markets: Russia, Mexico, Belgium demonstrate smaller but notable investment presence

Heatmap Patterns: Global Investors concentrate in high-value US regions; First-Time Buyers more dispersed internationally.

5.3 Investment Behavior Analysis

Box Plot Analysis: Global Investors show the widest investment spread ($500K–$3.5M), indicating internal heterogeneity despite cluster membership

Violin Plot (Age): All segments center around 55–60 years, but First-Time Buyers show a younger tail (30–45)

Loan Behavior: First-Time Buyers are loan-dependent (62% application rate); Corporate and Luxury segments predominantly self-finance

Purpose by Segment: Global Investors split ~70/30 Home/Investment; Corporate Buyers heavily investment-focused.

5.4 Model Performance Metrics

Metric	Value	Interpretation
Silhouette Score	0.51	Moderate-to-good separation
Davies-Bouldin Index	0.62	Strong cluster cohesion
Elbow Point	k = 4	Optimal cluster count confirmed
PCA Explained Variance (2D)	~68%	Majority of variance captured in visualization

7. Conclusion

This research successfully demonstrates that unsupervised machine learning can reveal actionable buyer segments in real estate data. Four distinct segments were identified, with Global Investors representing the dominant but internally diverse majority. The combination of K-Means clustering, rigorous validation metrics, and interactive deployment through Streamlit provides PARCL with a sustainable competitive advantage in customer intelligence.

The dashboard transforms static segmentation into a living analytical tool, enabling continuous exploration, new client prediction, and data-driven decision making across marketing, sales, and investment functions.

8. References

MacQueen, J. (1967). Some methods for classification and analysis of multivariate observations. Proceedings of the Fifth Berkeley Symposium on Mathematical Statistics and Probability, 1(14), 281–297.

Rousseeuw, P. J. (1987). Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. Journal of Computational and Applied Mathematics, 20, 53–65.

Davies, D. L., & Bouldin, D. W. (1979). A cluster separation measure. IEEE Transactions on Pattern Analysis and Machine Intelligence, PAMI-1(2), 224–227.

Pedregosa, F., et al. (2011). Scikit-learn: Machine learning in Python. Journal of Machine Learning Research, 12, 2825–2830.

Streamlit Inc. (2024). Streamlit Documentation. Retrieved from https://docs.streamlit.io/

PARCL Co. Limited & Unified Mentor. (2026). Machine Learning Internship Program: Real Estate Buyer Segmentation Project Brief.